Getting Started with NVIDIA DLSS 4 for Unreal Engine

(Supported Engines: 5.7, 5.6, 5.5. Deprecated Engine: 5.4)

Last Updated: v8.5.0 February 2026

General Overview

DLSS 4 neural rendering technologies include DLSS Multi Frame Generation, DLSS Frame Generation, DLSS Ray Reconstruction, DLSS Super Resolution, Deep Learning Anti-Aliasing and others.

- **DLSS Multi Frame Generation (MFG)** boosts frame rates by using AI to generate up to three frames per rendered frame, while maintaining responsiveness with NVIDIA Reflex. Blackwell's 5th generation Tensor Cores and display engine with improved flip metering deliver the AI horsepower and optimal frame pacing for a seamless gaming experience.
- **DLSS Frame Generation (FG)** boosts performance by using AI to generate frames while maintaining great responsiveness with NVIDIA Reflex for GeForce RTX 40 Series GPUs.
- **DLSS Ray Reconstruction (RR)** enhances image quality by using AI to generate additional pixels for intensive ray-traced scenes. DLSS replaces hand-tuned denoisers with an NVIDIA supercomputer-trained AI network that generates higher-quality pixels between sampled rays.
- **DLSS Super Resolution (SR)** boosts performance by using AI to output higher-resolution frames from a lower-resolution input. DLSS samples multiple lower-resolution images and uses motion data and feedback from prior frames to construct high-quality images.
- **Deep Learning Anti-Aliasing (DLAA)** provides higher image quality with an AI-based anti-aliasing technique. DLAA uses the same Super Resolution technology developed for DLSS, constructing a more realistic, high-quality image at native resolution.
- **RTX Dynamic Vibrance (RTXDV)** enhances visual clarity by dynamically tuning color, saturation and contrast.
- **NVIDIA Image Scaling (NIS)** boosts frame rates using GPU scaling and sharpening.
- **NVIDIA Reflex Low Latency** reduces PC latency and increases responsiveness.
- **DLSS Movie Render Queue** adds DLSS/DLAA support to Movie Render Queue.

DLSS Super Resolution, Ray Reconstruction, and DLAA now use a new transformer AI model trained by NVIDIA's supercomputers for improved stability from one frame to the next, enhanced lighting detail, and more detail in motion.

System Requirements

Operating Systems	Windows 10, 11 64-bit
Dependencies	NVIDIA RTX GPU 572.16 driver or newer DX12 (for DLSS Frame Generation and Multi Frame Generation Features), DX11, Vulkan
Development Environment	Visual Studio setup for Unreal Engine development.

Where to find everything

Here's an outline of everything you downloaded in the DLSS 4 zip file.

Quick Start Guide.pdf: That's this document. It is your primer/explainer on all contents of DLSS 4 and how to install the plugins. Please refer to the plugin Documentation for further details.

ReleaseNotes_DLSS_Streamline_UE_plugins.pdf: This document is for any known issues, bugs, or other details worth noting to developers. These are issues and bugs we are tracking internally and intend to resolve in future releases.

Samples folder: Contains the DLSS 4 Sample project. This project uses every feature of DLSS 4 in raster and ray tracing modes and can be a quick and easy way to validate DLSS 4 is working correctly for you. It also has a clean and well-structured BP UI you can use and modify for your project.

Plugins folder: Contains all of the DLSS 4 plugins for UE developers. These will be the plugins you install using the installation steps below. Each feature is implemented using a subset of those plugins:

Feature	**Plugins folder and dependencies**
DLSS Super Resolution (DLSS-SR) Deep Learning Anti-Aliasing (DLAA) DLSS Ray Reconstruction (DLSS-RR)	Plugins/DLSS Plugins/StreamlineNGXCommon
Movie Render Queue support for DLSS-SR,DLSS-RR, DLAA	Plugins/DLSSMoviePipelineSupport
DLSS Frame Generation (DLSS-FG) DLSS Multi Frame Generation (DLSS-MFG)	Plugins/StreamlineDLSSG Plugins/StreamlineReflex Plugins/StreamlineCore Plugins/StreamlineNGXCommon
NVIDIA RTX Dynamic Vibrance (also called DeepDVC)	Plugins/StreamlineDeepDVC Plugins/StreamlineCore Plugins/StreamlineNGXCommon
NVIDIA Image Scaling and Sharpening (NIS)	Plugins/NIS
NVIDIA Reflex Low latency	Plugins/StreamlineReflex Plugins/StreamlineCore Plugins/StreamlineNGXCommon

Platforms folder: Contains the WinGDK plugin extensions. Install the relevant plugin extensions if you intend to ship your title on the Microsoft GamePass platform. Please refer to the respective plugin documentation on how to do that:

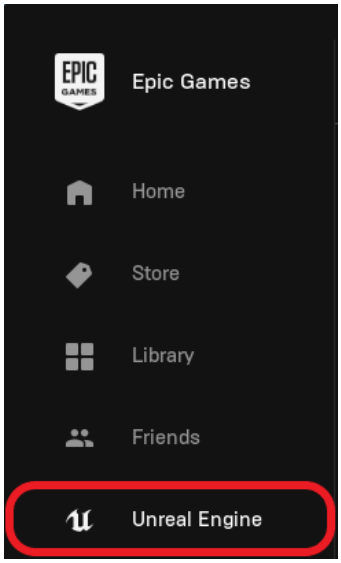
Plugin	**WinGDK plugin extensions**
Plugins/DLSS	Platforms/WinGDK/Plugins/DLSS/DLSS_WinGDK.uplugin
Plugins/StreamlineCore	Platforms/WinGDK/Plugins/StreamlineCore/StreamlineCore_WinGDK.uplugin
Plugins/*any other*	No plugin extension required for any of the other plugins

Documentation folder: Contains all of the PDF documents that cover DLSS 4 technology and their uses. Some of these documents are specifically intended for the SDK version of DLSS, but they can still be handy programming references if you are working with code. Below are all the included documents, grouped by large topics:

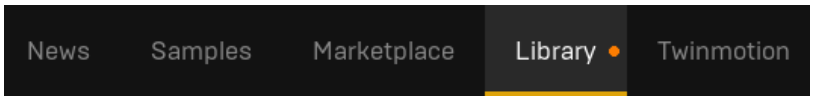
- **Application designers, focus on user interface, game settings**
  - **RTX UI Developer Guidelines**: The user interface guideline document that proposes best practices and standard usage for DLSS 4 features. This document makes a number of suggestions for best practices to the application developer.
  - **RTX UI Developer Guidelines Chinese Version**: Chinese language version of the above document.
- **Application designers, focus on blueprints, programmers focusing on rendering**
  - **DLSS_Frame_Generation_Reflex_DynamicVibrance_UE_Plugins_Guide**: An Unreal Engine specific document to help guide you through issues and troubleshooting FG, MFG, Reflex and RTXDV.
  - **NIS_UE_Plugin_Guide**: An Unreal Engine specific document to help you with all the relevant features and uses of NIS.
  - **DLSS_Super_Resolution_UE_Plugin_Guide**: An Unreal Engine specific document that covers DLSS SR, DLAA and RR most specifically. This is the document to look at if you want to go deeper on upscaling and denoising issues.
- **Programmers, with focus on the underlying Streamline and NGX SDKs**
  - **Streamline Developer Guide**: This document is mainly a reference for the SDK version of Streamline. Streamline is a bundling/wrapping technology used to take different technologies and link them together, making developer integration easier and more direct.
  - **DLSS_Programming_Guide_Release**: Mainly for the SDK version of DLSS but can still be a useful coding reference.
  - **DLSS-RR Integration Guide API**: A coding/API reference document for RR.
  - **NIS_SDK_Programming_Guide**: An SDK reference document for NIS.
- **Business Development, Legal**
  - **NVIDIA RTX SDK EULA**: The end user license agreement for DLSS 4 and related technologies. For any legal questions please refer to this document.

Installation Steps

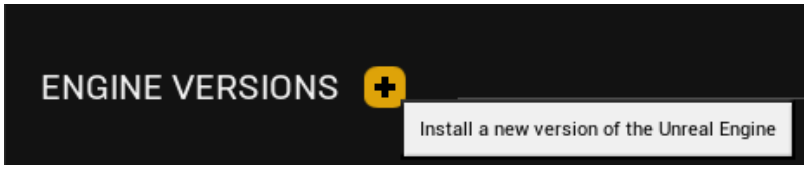
1. Install the Epic Games Launcher from Epic's website: https://www.epicgames.com/store/en-US/download
2. Log into your Epic Games account and navigate to the Unreal Engine tab to the left of the launcher.



3. At the top window toolbar, click `Library`



4. Install the latest available, or previous versions of Unreal Engine 5 by pressing the `+` button to the right of 'ENGINE VERSIONS`



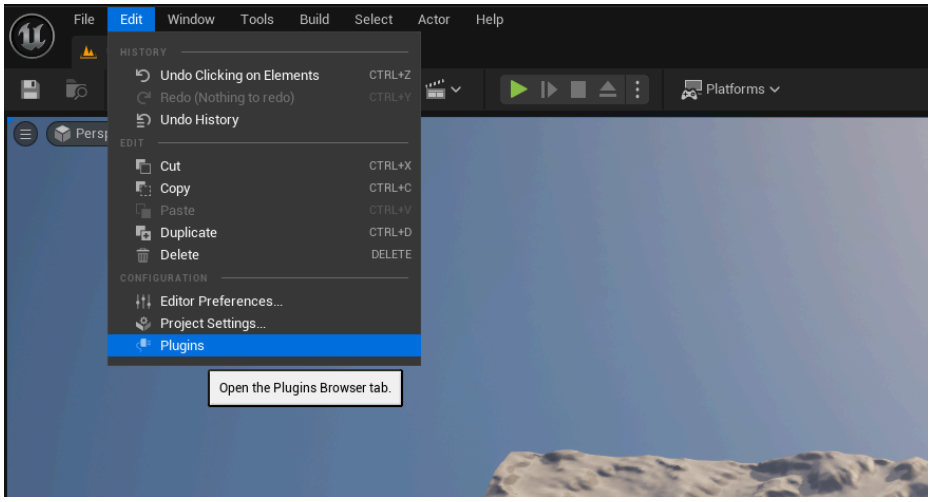
5. Unzip the DLSSPlugin.zip file downloaded from the NVIDIA DLSS Plugin download page.Once extracted, navigate to the “Plugins” folder where you will see the following:

Name	Type
DLSS	File folder
DLSSMoviePipelineSupport	File folder
NIS	File folder
Streamline	File folder
StreamlineCore	File folder
StreamlineDeepDVC	File folder
StreamlineDLSSG	File folder
StreamlineNGXCommon	File folder
StreamlineReflex	File folder

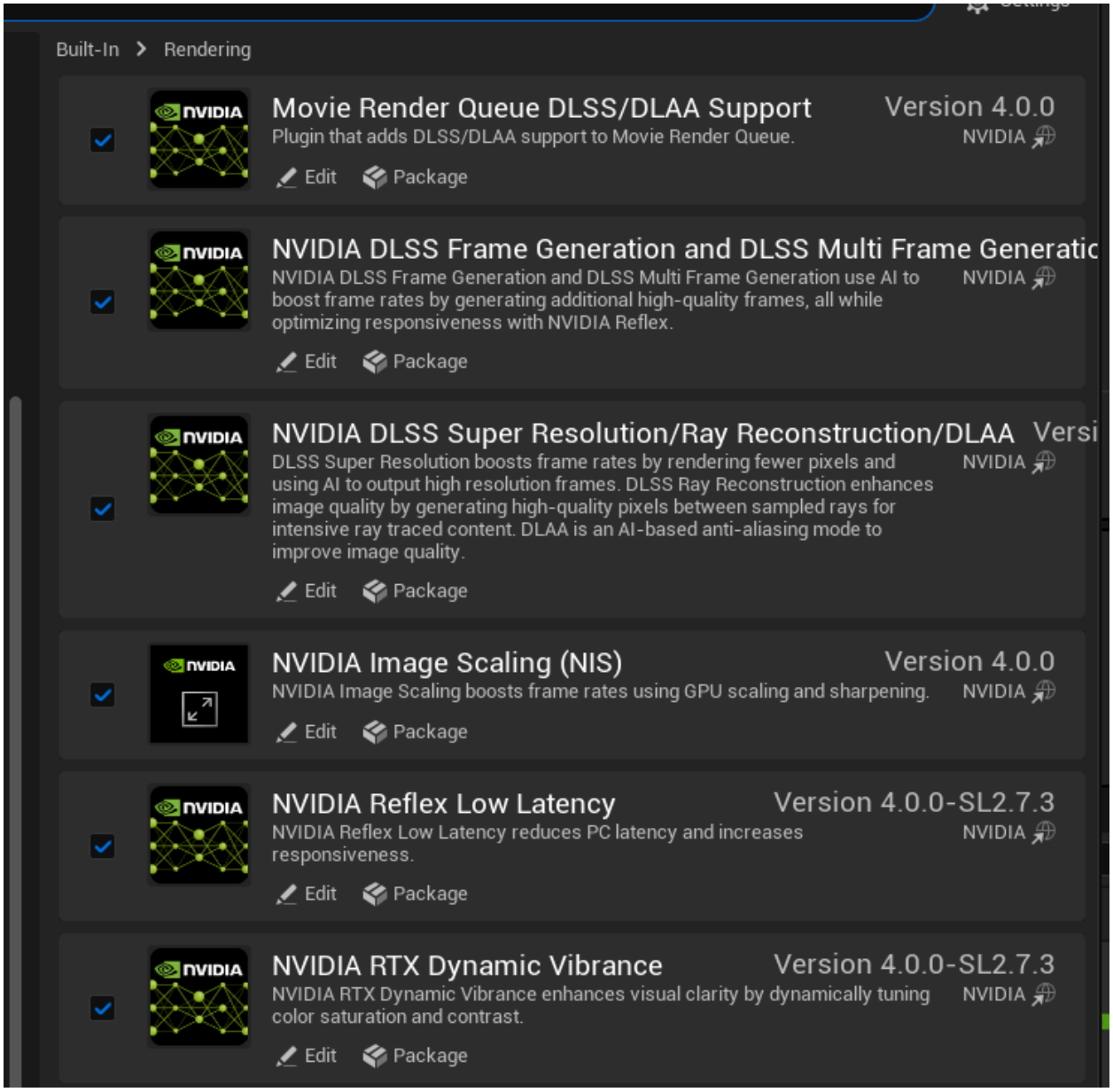
6. Place the unzipped plugin folders in “*your engine version*”\Engine\Plugins\Marketplace
    a. There are a total of 8 DLSS related plugin folders as listed above.
7. Launch Unreal Engine 5.

Note: If this is your first time running the editor, it will require you to install the prerequisites. It may also take longer to load up than usual.

8. Once you’ve opened up your project, go to the toolbar at the top of the screen and click `Edit`. Then, in the dropdown menu, select `Plugins` to open up the Plugins browser.



9. Search for “NVIDIA | DLSS” in the plugins search bar (with the vertical slash between search terms). This will give you a complete list of all NVIDIA or DLSS related plugins.

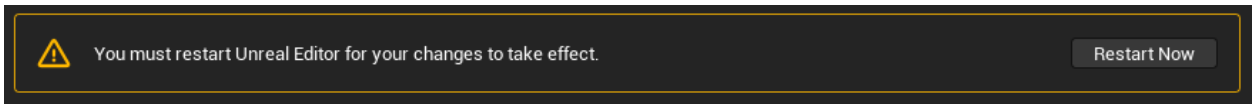


10. Check the `Enabled` box to the left of the NVIDIA DLSS icons.
    a. There are a total of 8 plugins related to DLSS 4. Not every plugin is required or necessary, this will depend on the technology that is important to your project.
    b. The Streamline (deprecated) plugin is only necessary if you are upgrading from an older version of DLSS/Streamline and can be ignored for new projects.
    c. RTX Dynamic Vibrance for example, is optional and should only be enabled if you intend to use it. Same goes for the Movie Render Que (MRQ) DLSS plugin.
    d. NVIDIA Image Scaling (NIS) serves two purposes, it is the sharpener for DLSS and is a broadly compatible upscaler for non-RTX hardware. While not absolutely necessary for DLSS to function, if you wish to give users a sharpening

slider in the UI when DLSS is enabled (per our UI guidelines) then it is strongly recommended you enable NIS for this reason. Additionally, it can be a useful cross-platform fallback upscaler and sharpener when DLSS is not available.

e. NVIDIA Reflex Low Latency 4 plugin replaces the older Reflex 1.6 plugin built-in to Unreal Engine and should be used instead of 1.6. Make sure to disable the older plugin.

11. A new prompt will appear at the bottom of the plugins window indicating you must restart your editor each time a new plugin is enabled.



12. Restart the editor.
13. Navigate to the “Documentation” folder for more detailed instructions for each of the DLSS technologies.

If you are experiencing issues in getting any of the NVIDIA Plugins enabled, please refer to our Troubleshooting Tips & Tricks Guide. For additional support, please contact us at dlss-support@nvidia.com.